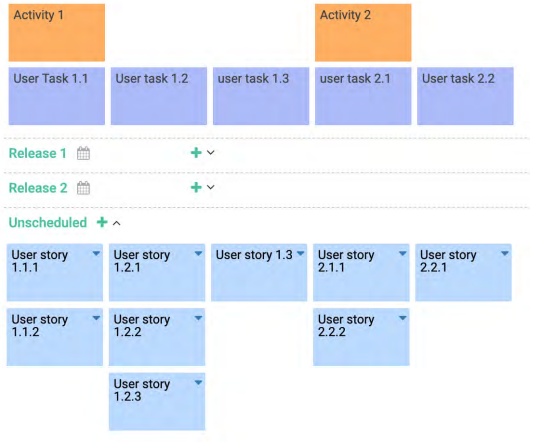


ΕΘΝΙΚΟ ΜΕΤΣΟΒΙΟ ΠΟΛΥΤΕΧΝΕΙΟ
Σχολή Ηλεκτρολόγων Μηχανικών και Μηχανικών Υπολογιστών
Τεχνολογίες Υπηρεσιών Λογισμικού

Τεχνολογίες Υπηρεσιών Λογισμικού

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Behavior-Driven Design

How to effectivelly(?) apply "*Agile*" in design (+ planning)

Agile is useful when...

Development is done closely with stakeholders

- Users, customers, developers, operators, integrators, ...

A working prototype is available to stakeholders

- Improve over iterations
- Each iteration takes 1 – 2 weeks

Confirm you are doing the right thing

- Doing it right, is another discussion

Behavior-Driven Design

Concepts

- Discuss the behavior of the app with the customer(s) before and during development, to create user stories
- Focus on hehavior, not implementation

- Use short descriptions: **title, who, what, why**
- "As a visitor, I want to see examples of available quizzes, in order to decide if I will enrol"

User stories

User stories: easier said than done (correctly)...

- What is a good user story?
- How to create with the customer?
- How to estimate difficulty (effort)?
- How to prioritize?
- How to track who does what?
- How to verify code that implements stories?

User stories

A user story should be SMART:

- Specific (no generic wording)
- Measurable (what is the measure?)
- Achievable (in 1 iteration!)
- Relevant (does it make sense?)
- Timeboxed (finish or quit)

Good user stories: specific

Understand

- Inputs, including pre-conditions
- Outputs, including post-conditions
- Value added (logic to implement)

Example: display a line diagram

- *Given* that the data connection is active
- *When* I select dates and press "refresh"
- *Then* the diagram should display on the screen

Good stories: measurable

What to measure: effort *vs* value of the result

- Effort is about resources (labour * time)
- VERY difficult to estimate in advance
- Even MORE so, if no clear understanding of what the story is about (= requirements) exists
- Even MORE so, if the story builds on other stories, half-completed

Good stories: achievable

General guidelines:

- Complete in one (1) iteration (usually sprint); If this cannot be done, break into smaller stories
- Deliver working code at the end (working **!=** dirty)

Nice to have:

- Experience
- Team maturity
- Good understanding of the problem

Good stories: relevant

Understand the customer's needs

- Ask questions (A.Fox suggests "5 whys")
- Challenge *your* understanding, not the customer's
- The team should be confident that they are doing the right thing (valid by the customer's standards)

Do not mix roles

- Develoment is one thing
- Consulting is another

Good stories: timeboxed

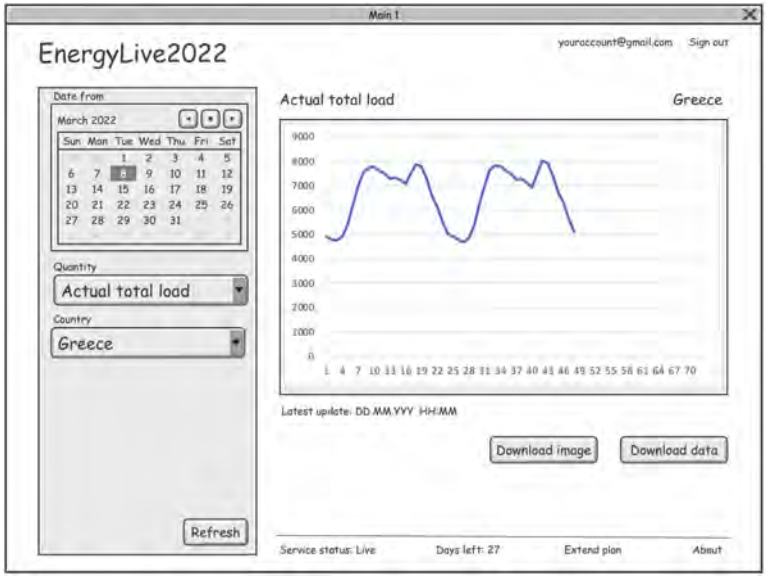
Stop when time or budget exceeded

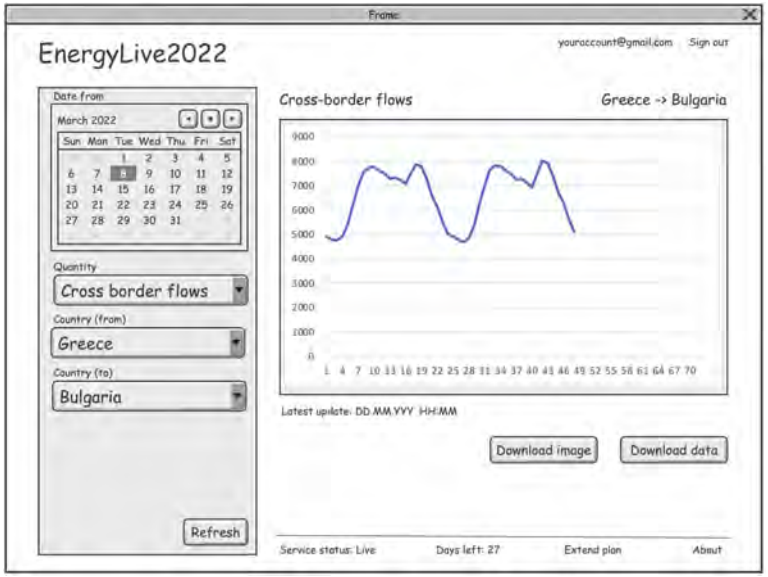
- Option 1: give up, roll back
- Option 2: break into smaller stories & reschedule

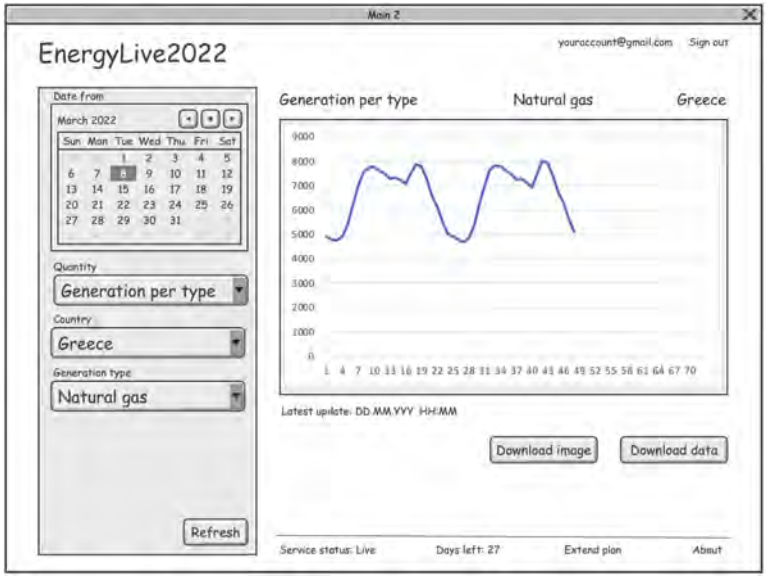
Agile is not forever

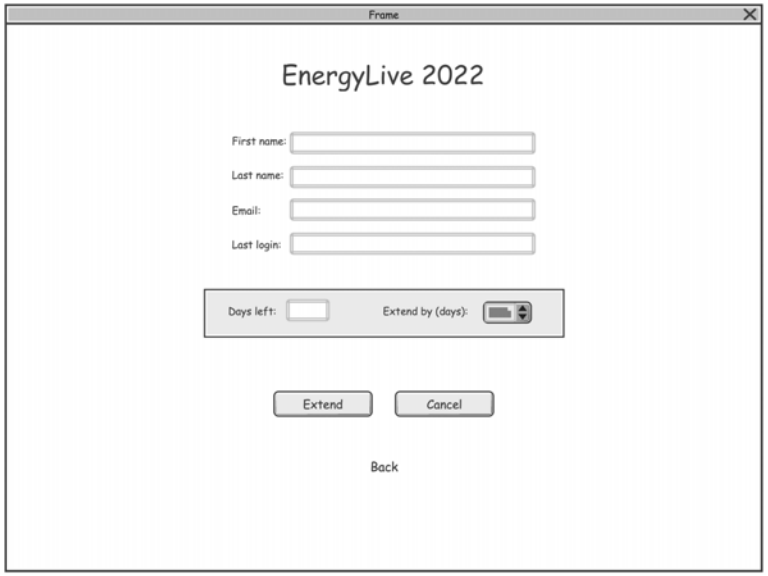
- Avoid underestimating the size of the project
- Use tools to support your work

User stories: UI storyboards

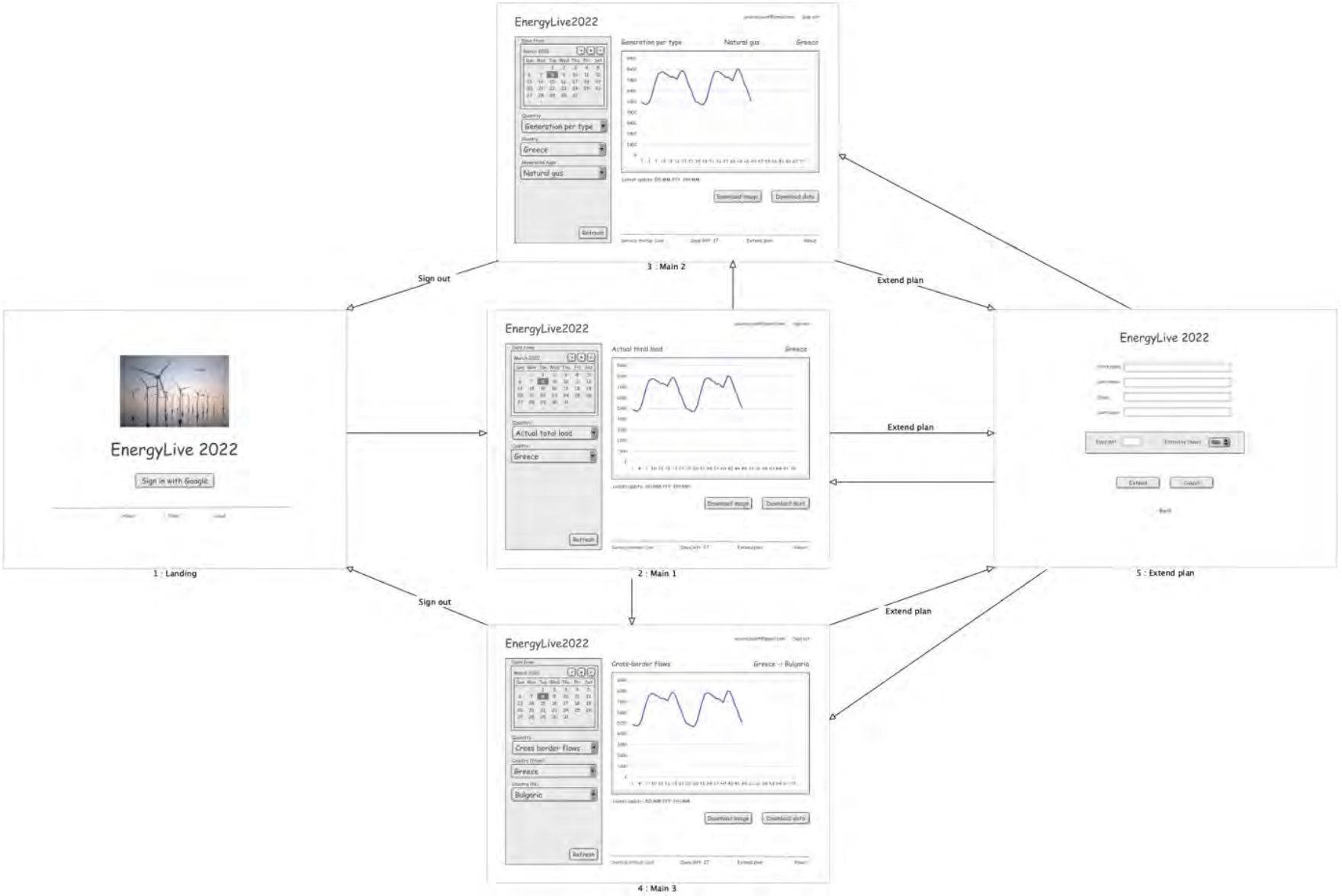




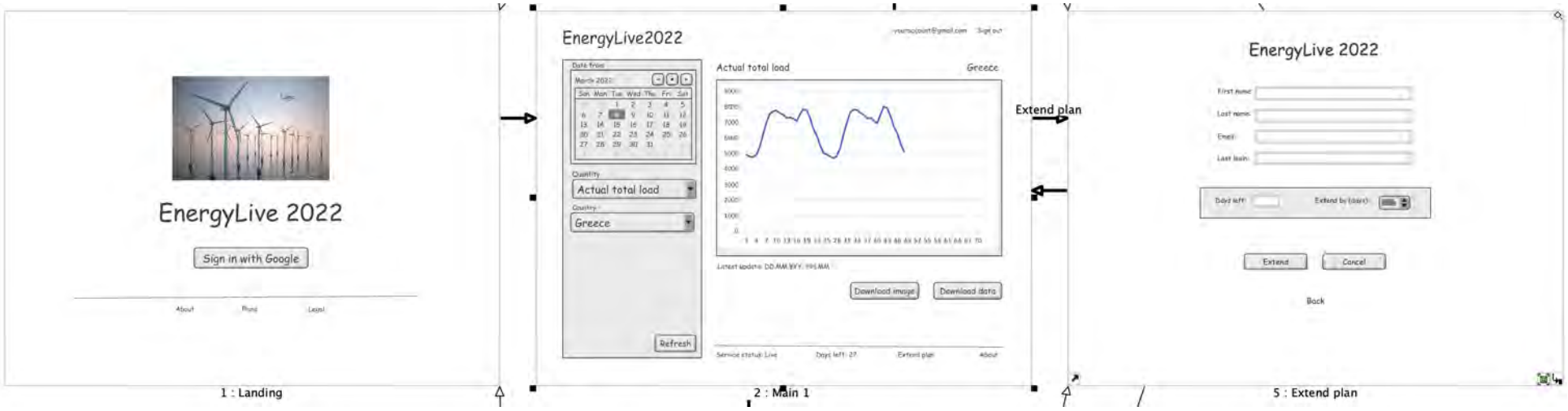




User stories: UI storyboards

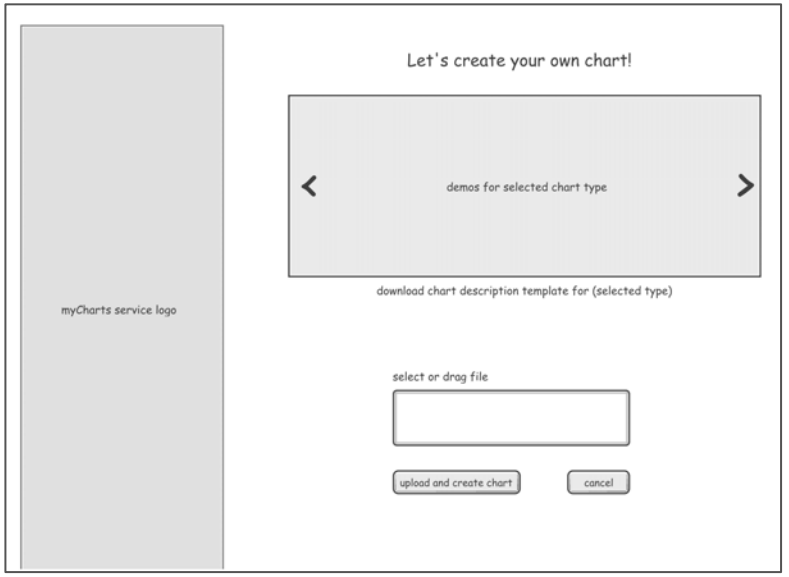


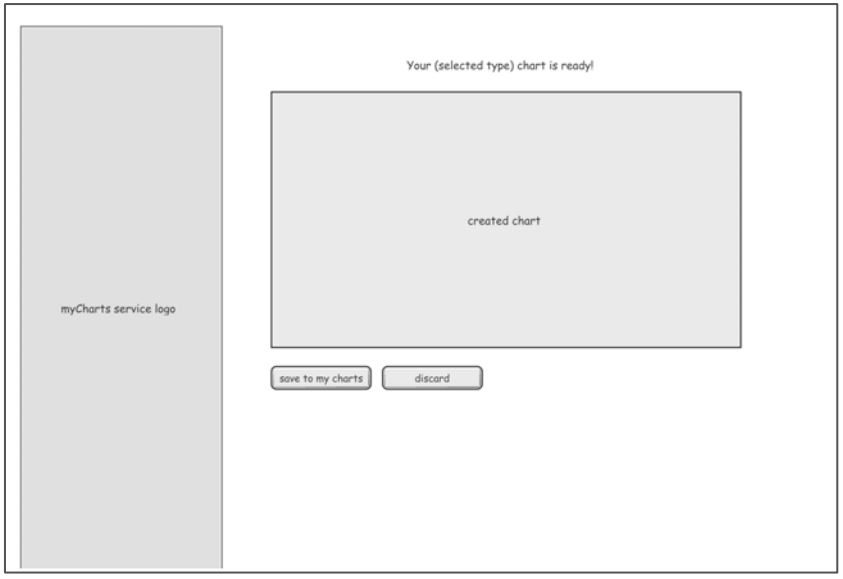
User stories: UI storyboards

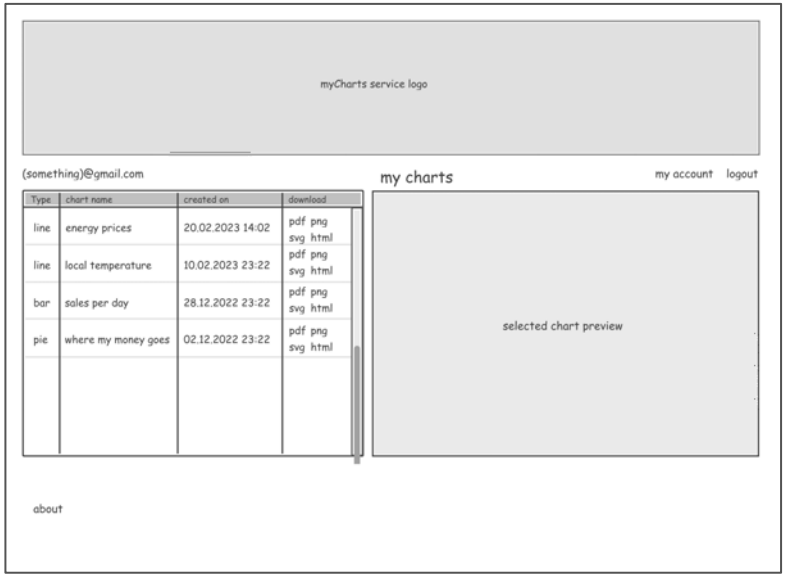


User stories: UI storyboards

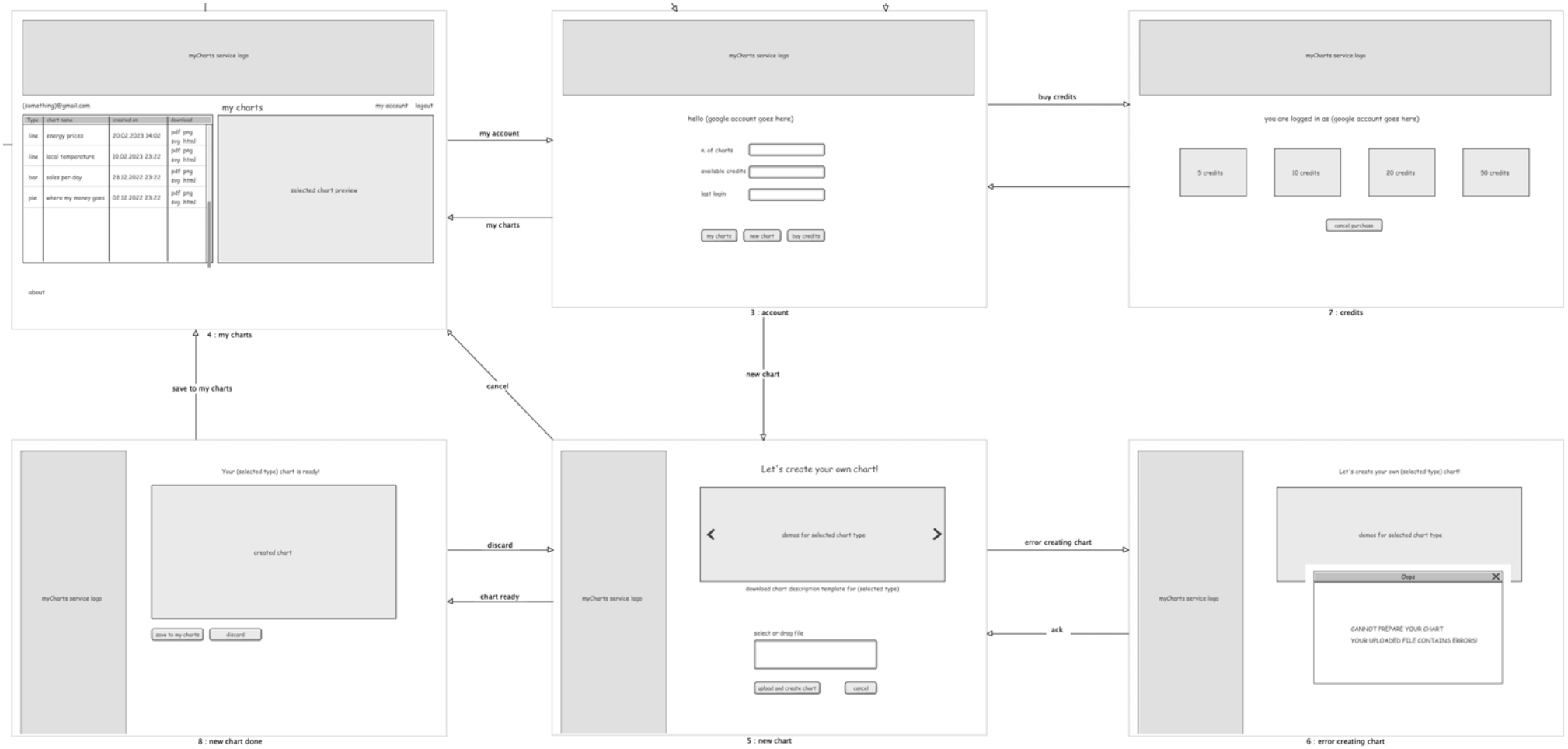




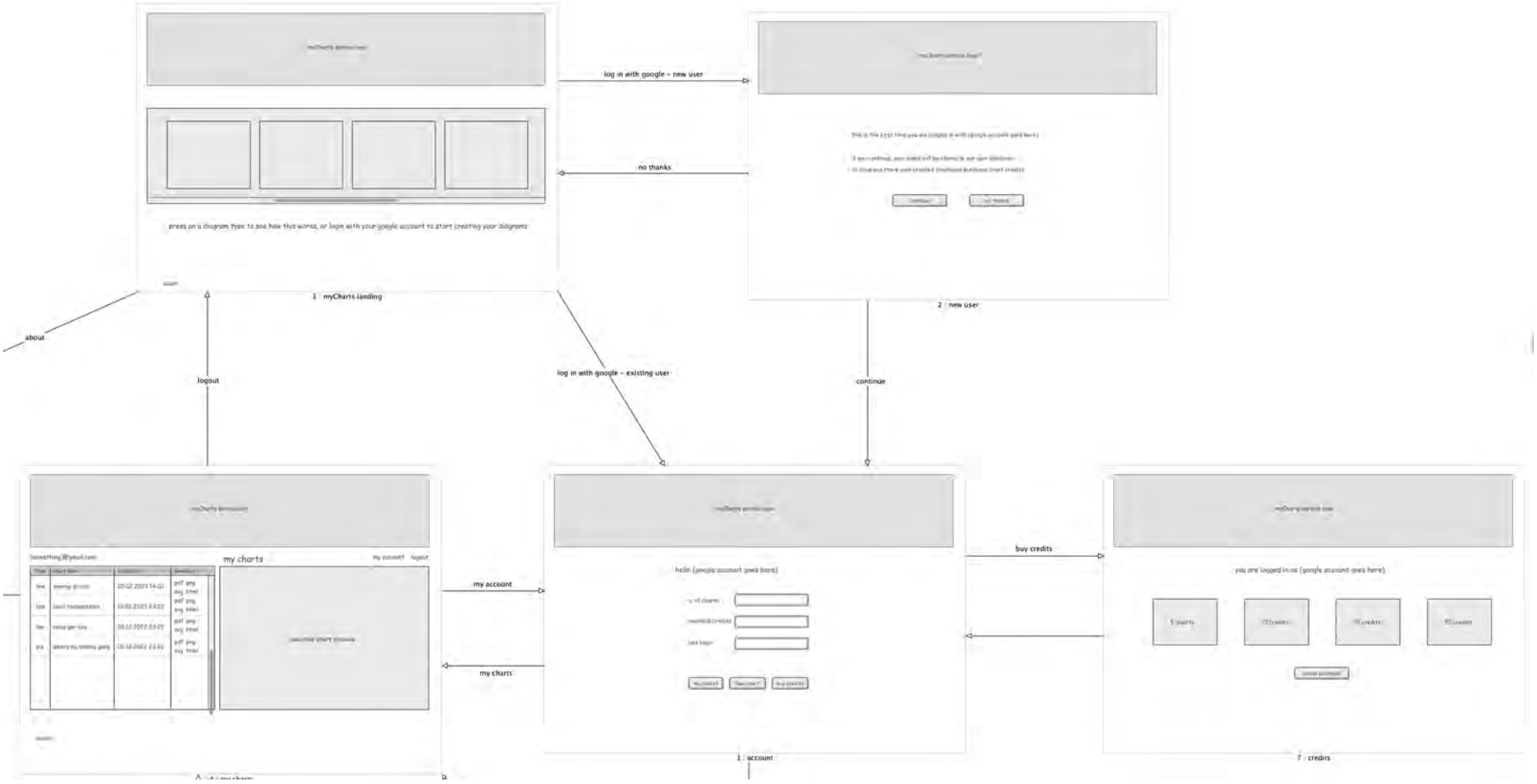




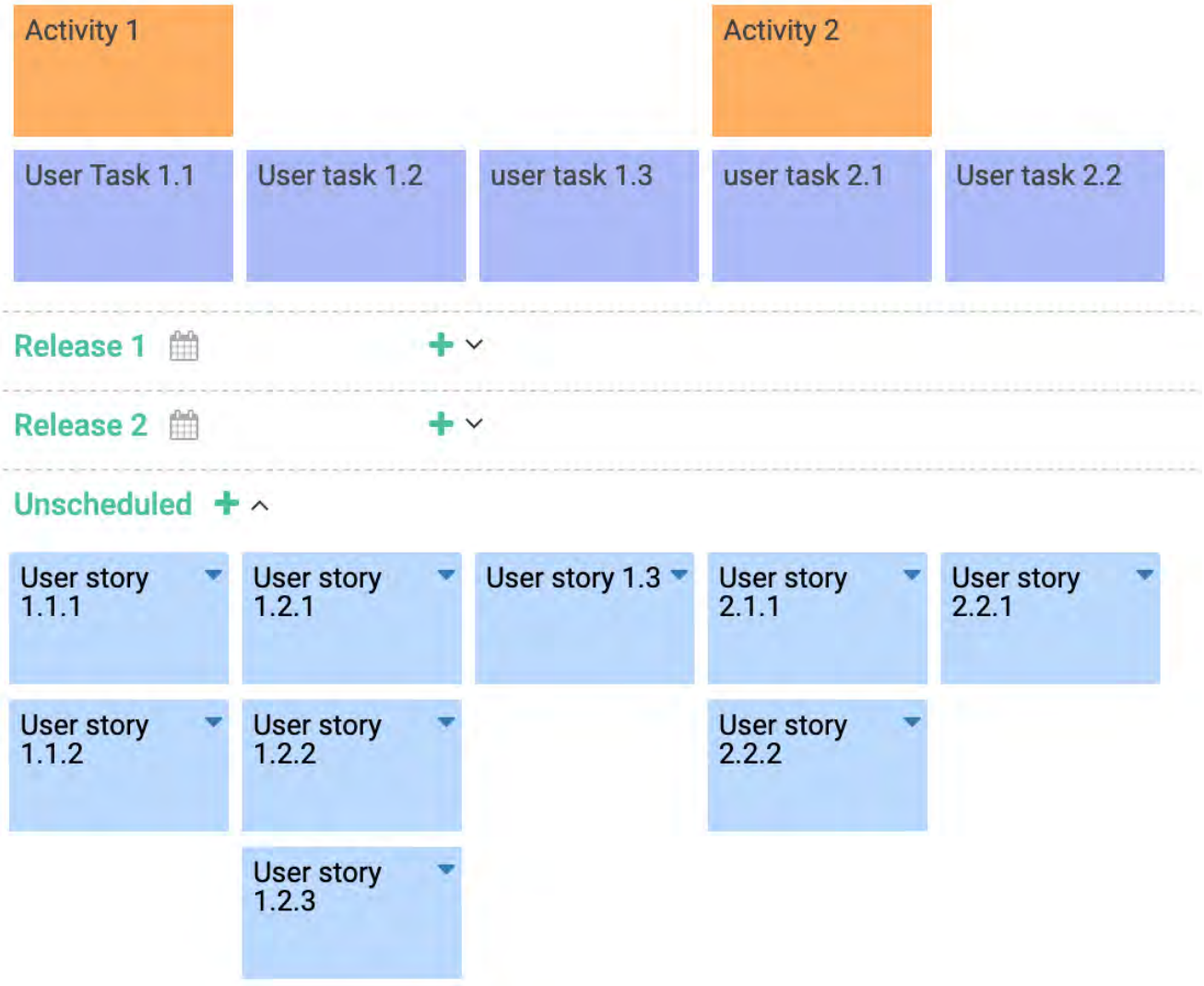
User stories: UI storyboards



User stories: UI storyboards



User stories: tools

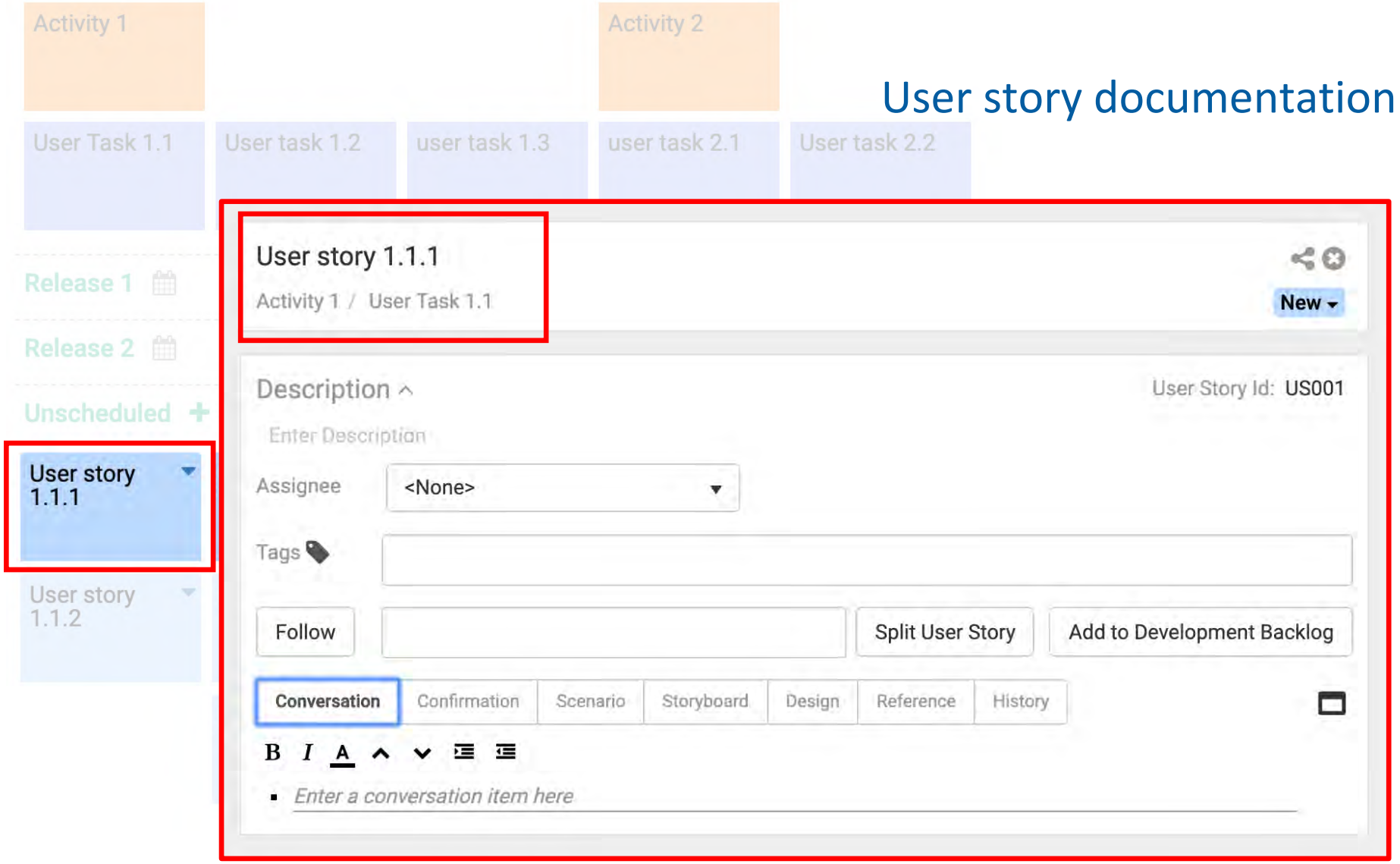


User activities

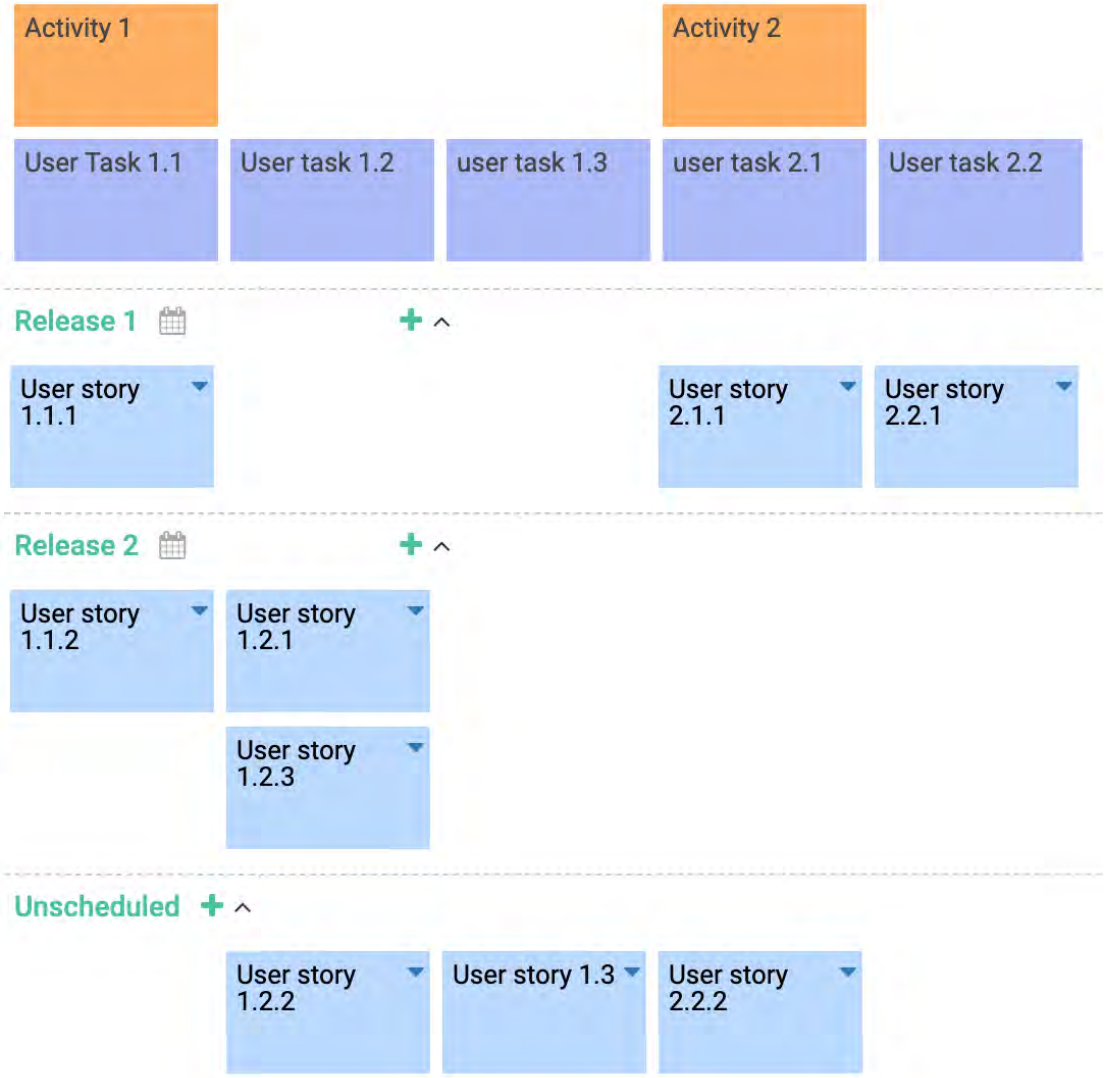
User tasks

User stories

User stories: tools

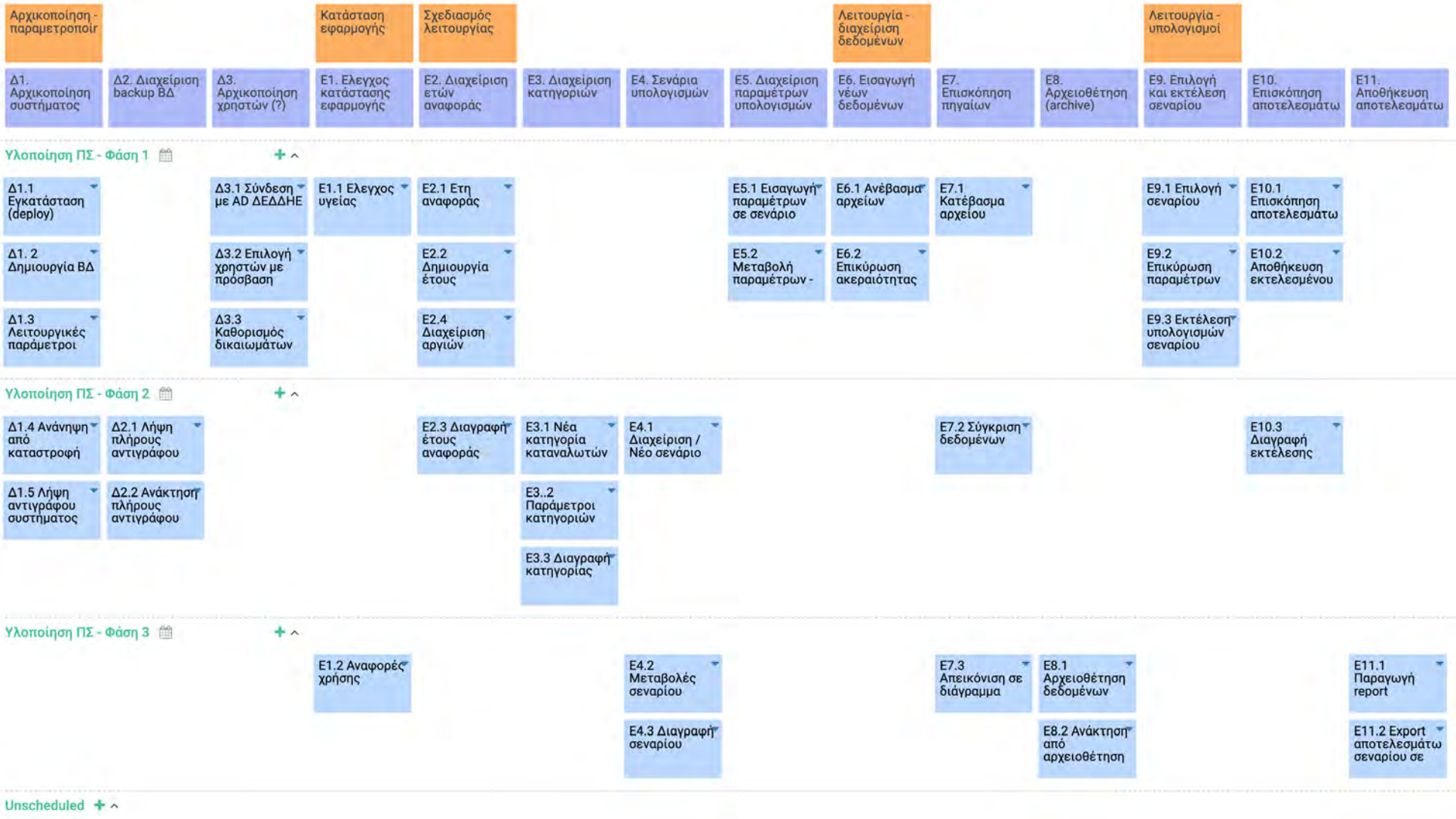


User stories: tools



Map to releases or sprints

User stories: real-world example



Example: course semester project 2020-2021

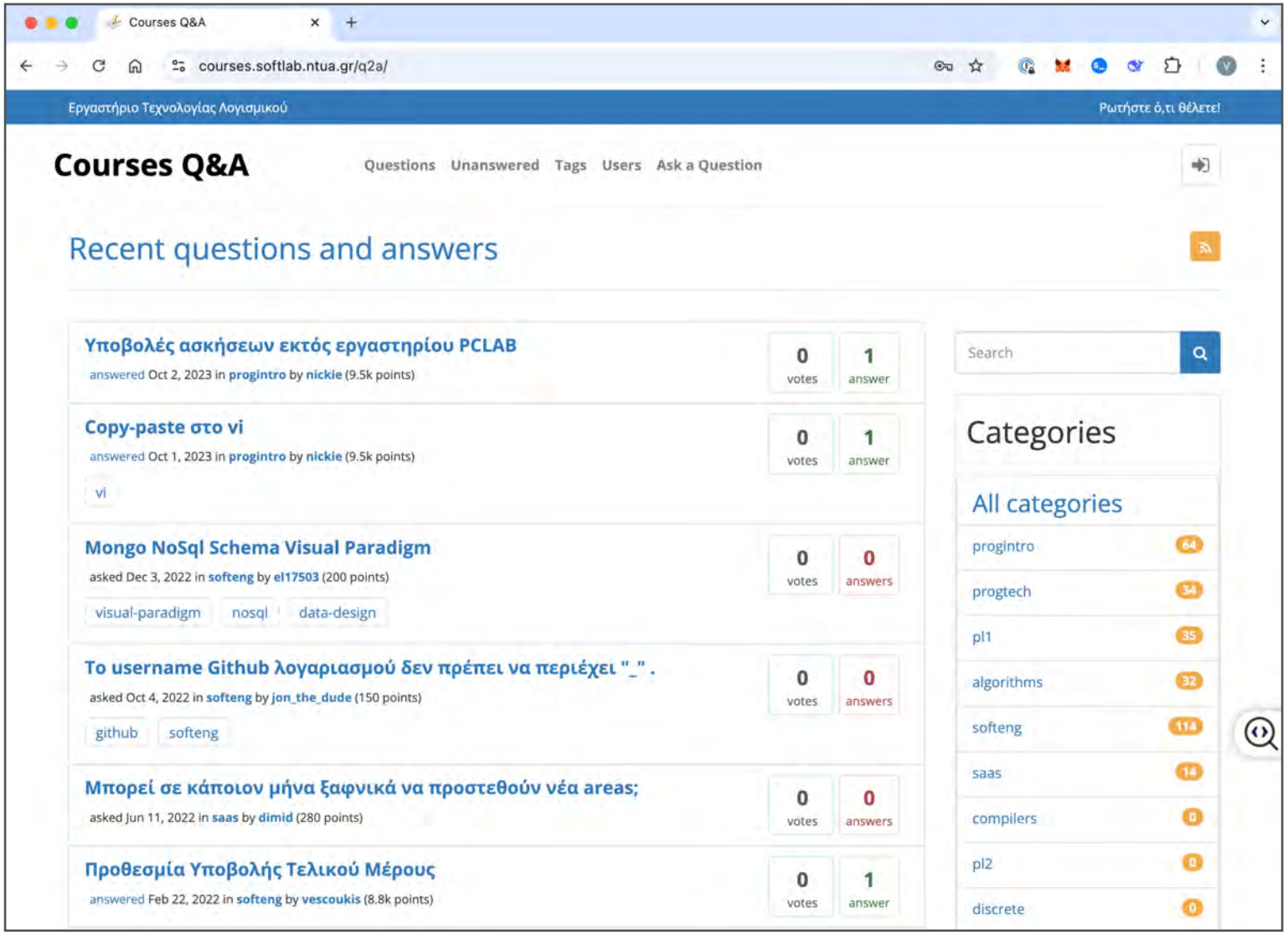
Askme**anything**

Το “**ask**me**anything**” είναι μια εφαρμογή ερωτήσεων - απαντήσεων η οποία θα προσφέρεται ως υπηρεσία λογισμικού (SaaS) σε υποθετικούς πελάτες όπως εταιρίες που προσφέρουν υπηρεσίες τύπου help desk, εταιρίες που περιγράφουν τις εσωτερικές τους διαδικασίες με τη μορφή ερωτήσεων-απαντήσεων, εκπαίδευση κ.ά. Στο πλαίσιο της εργασίας θα αναπτυχθεί μια απλουστευμένη εκδοχή της εφαρμογής η οποία εκτελεί τις ακόλουθες λειτουργίες:

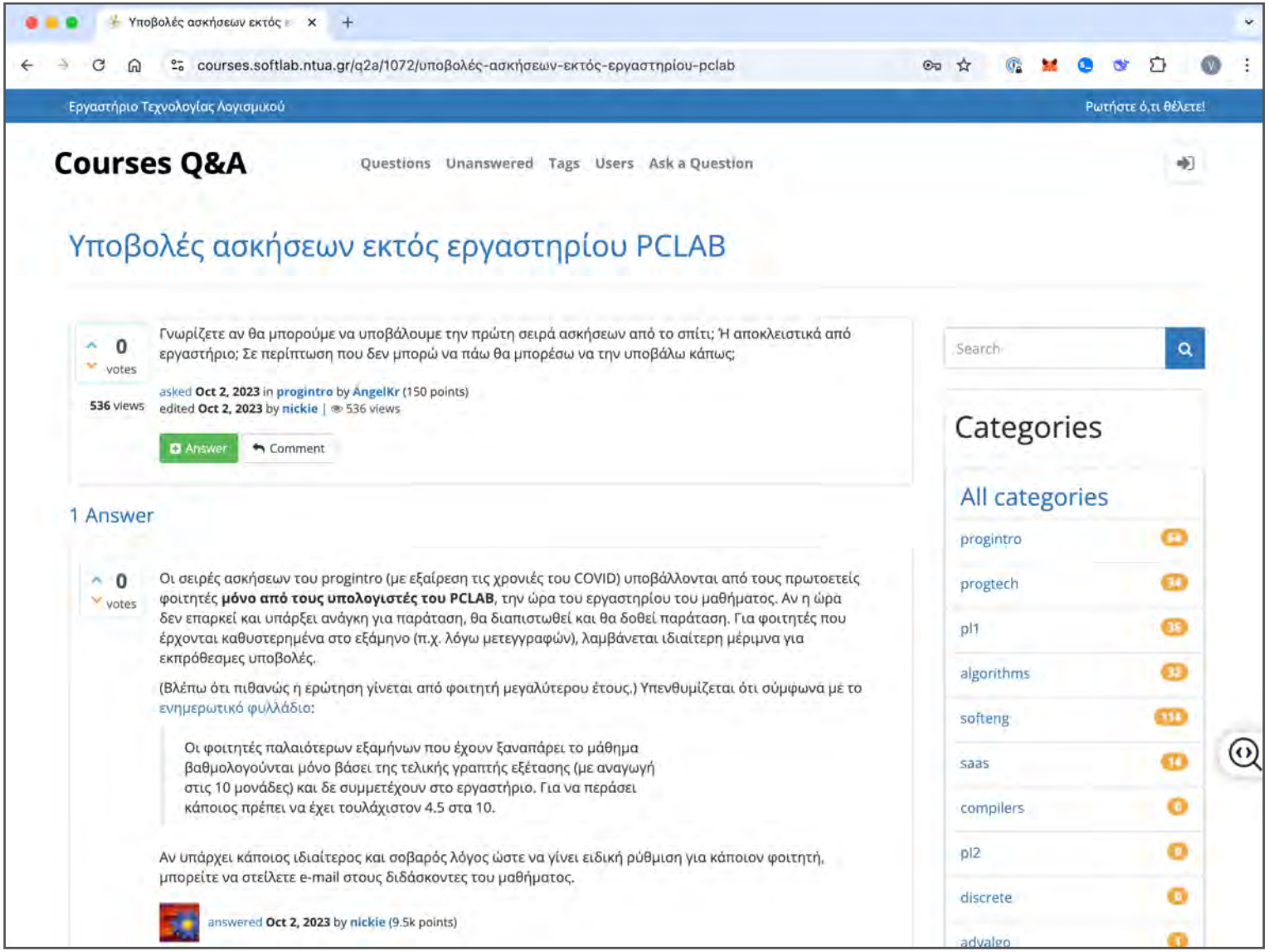
- Δημιουργία ερώτησης
- Χαρακτηρισμός με λέξεις-κλειδιά
- Απάντηση ερώτησης
- Περιήγηση σε ερωταπαντήσεις
- Στατιστικά ερωτήσεων με λέξεις-κλειδιά
- Στατιστικά ερωτήσεων με ημερομηνίες
- Sign up, sign in

Κάθε ομάδα θα υλοποιήσει **δύο εκδοχές** του **ask**me**anything**, οι οποίες θα υλοποιούν 2 από τις ακόλουθες αρχιτεκτονικές SOA, Microservices και MVC

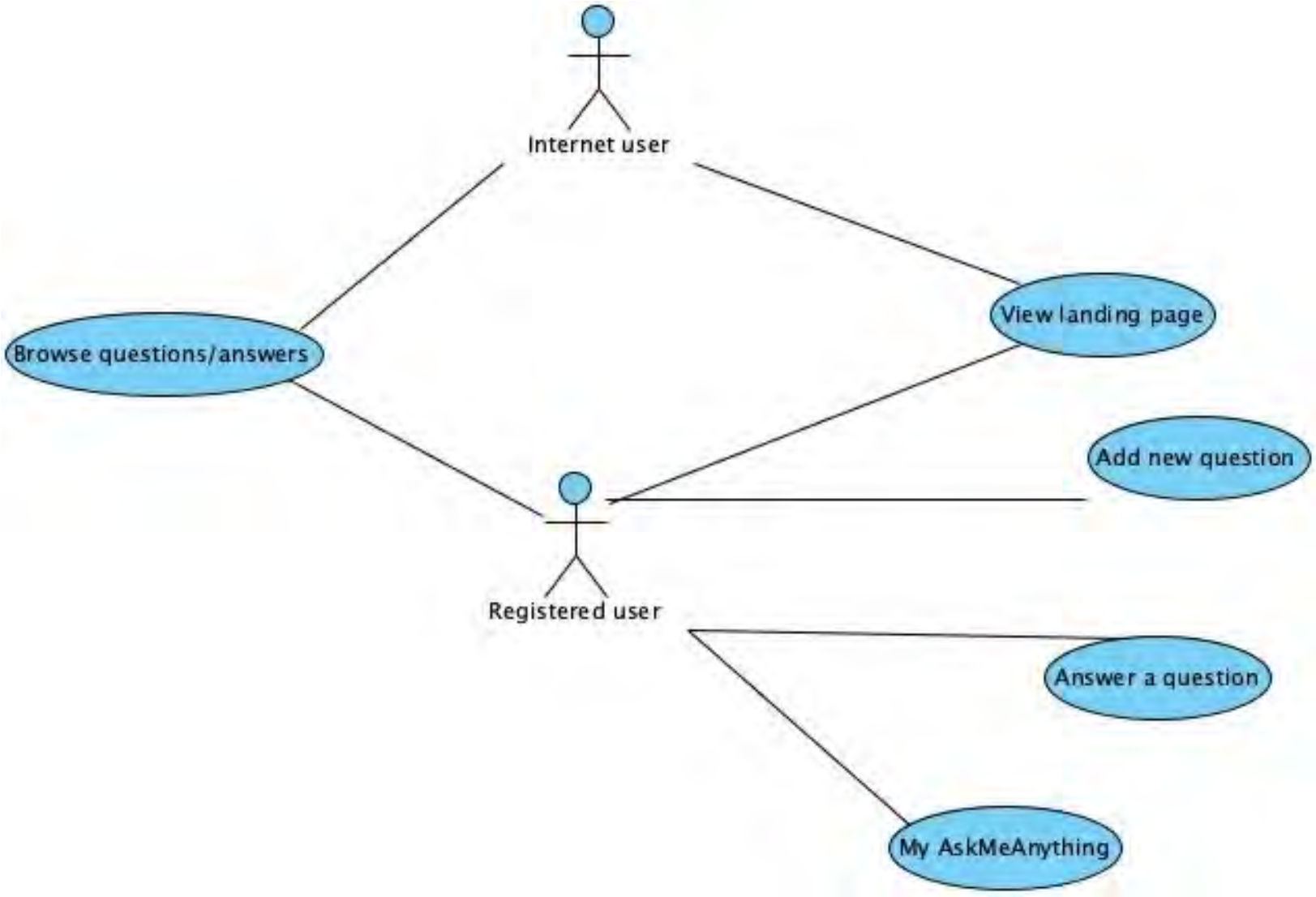
Askmeanything



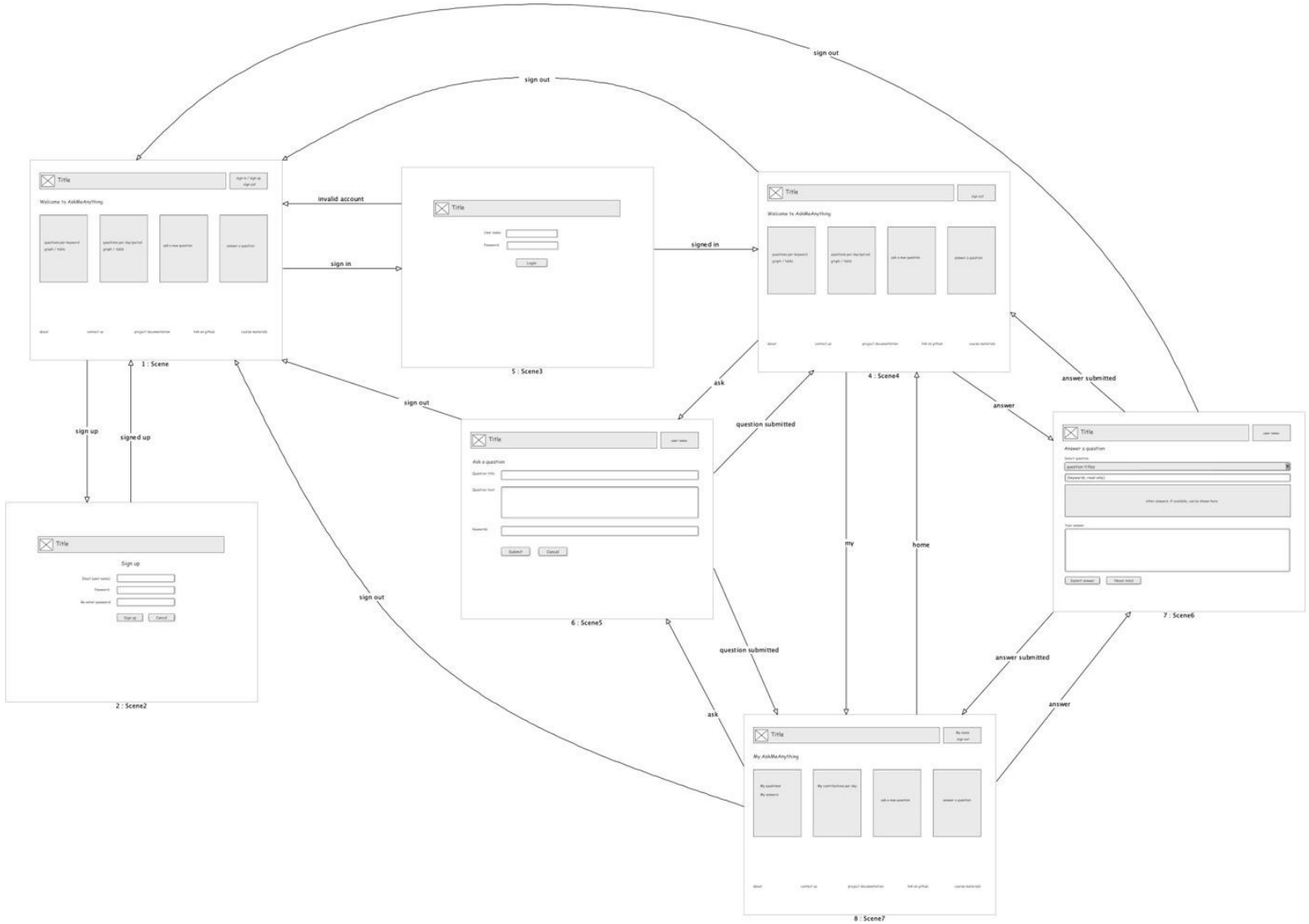
Askmeanything



Askmeanything



Askmeanything



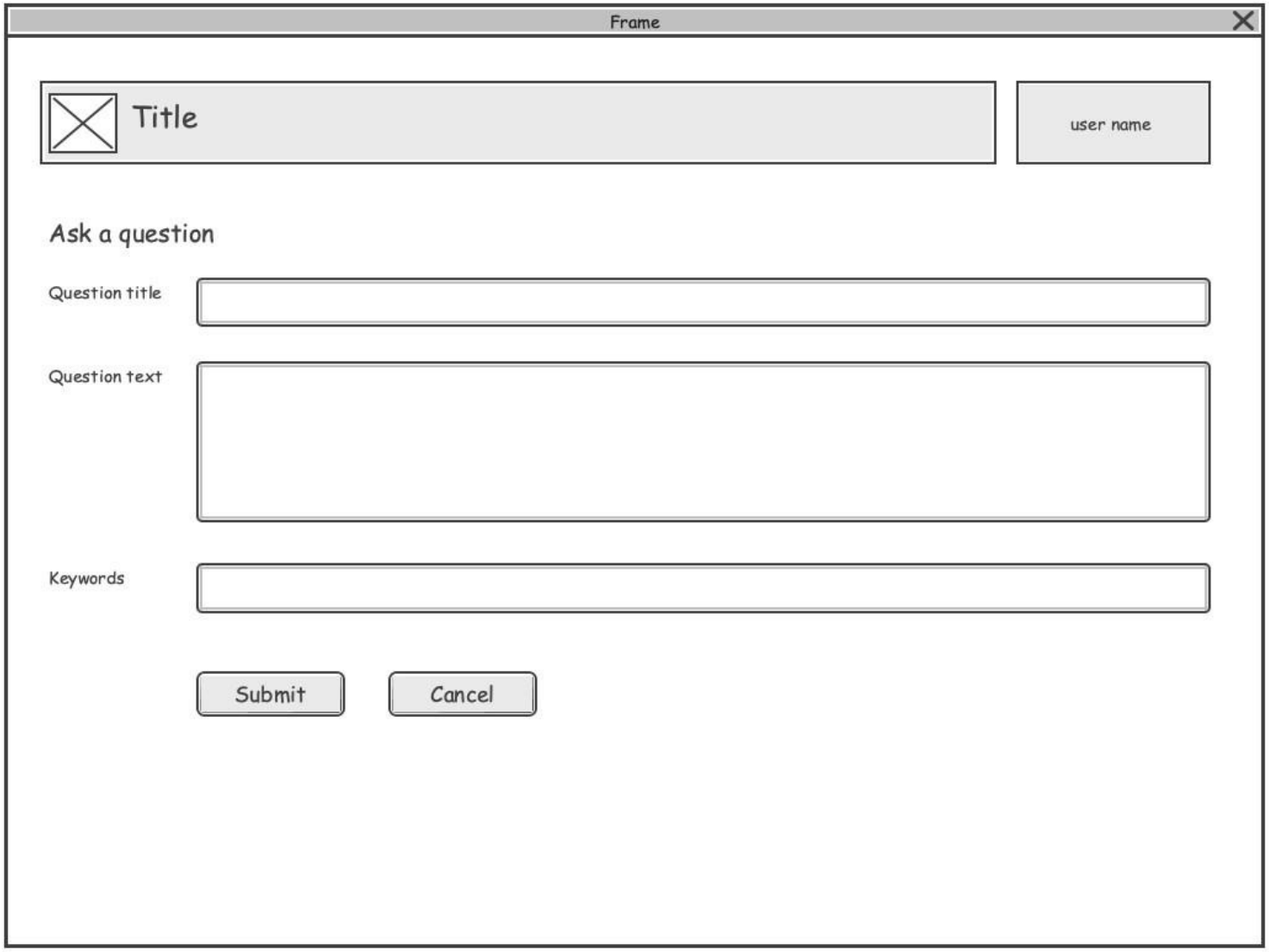
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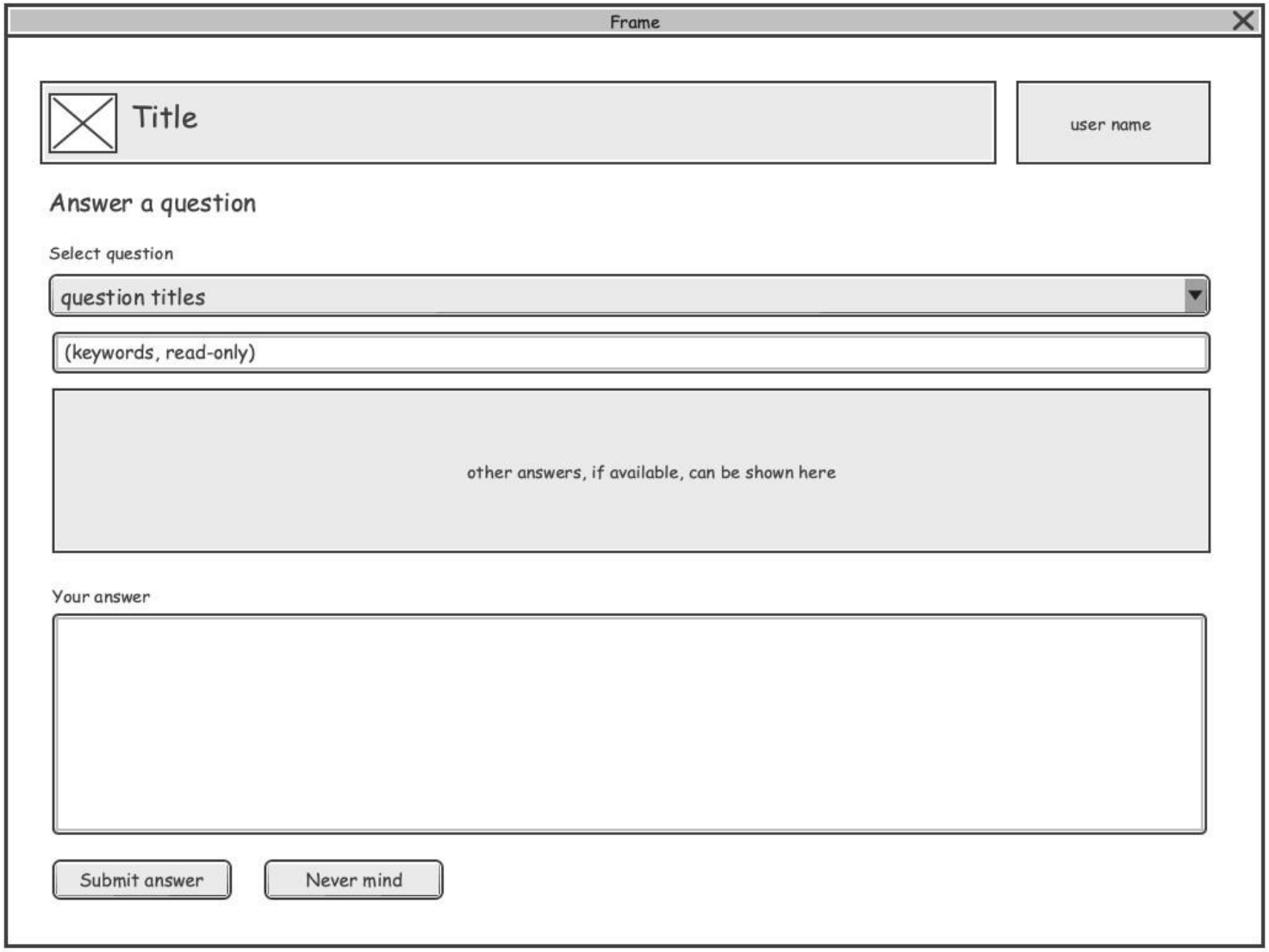
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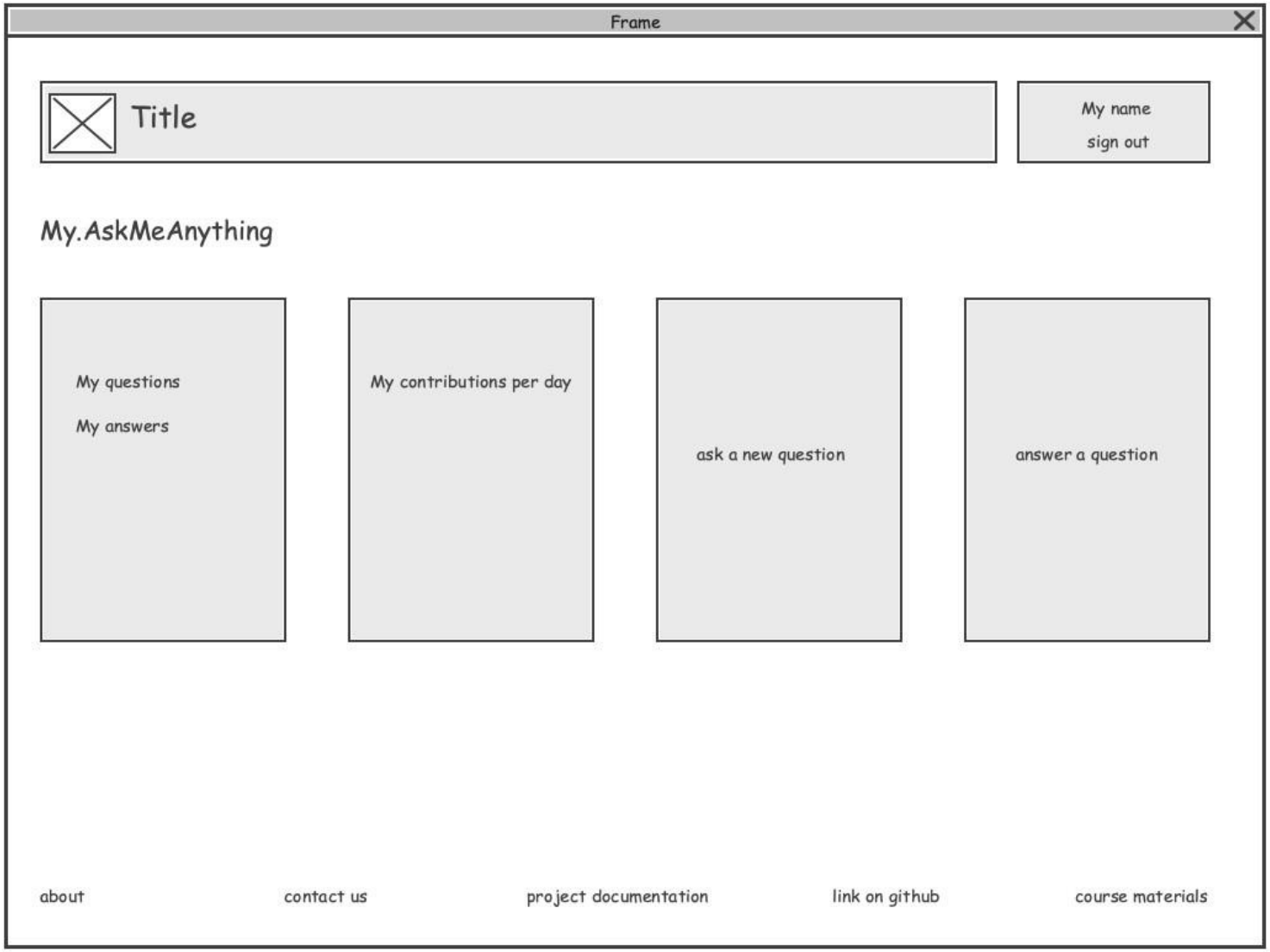
Askmeanything



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Askmeanything



Askmeanything

